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ORIGINAL INVESTIGATION

Time Course of Neuromuscular, Hormonal, and Perceptual Responses Following Moderate- and High-Load Resistance Priming Exercise

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Purpose: The aim of this study was to map responses over 32 hours following high-load (HL) and moderate-load (ML) half-squat priming. ***Methods:*** Fifteen participants completed control, HL (87% 1RM), and ML (65% 1RM) activities in randomized, counterbalanced order. Countermovement jump (CMJ), squat jump (SJ), saliva testosterone, saliva cortisol, and perceptual measures were assessed before and 5 minutes, 8 hours, 24 hours, and 32 hours after each activity. Results are presented as percentage change from baseline and 95% confidence interval (CI). Cliff delta was used to determine threshold for group changes. ***Results:*** SJ height increased by 4.5% (CI = 2.2–6.8, Cliff delta = 0.20) 8 hours following HL. CMJ and SJ improved by 6.1% (CI = 2.1–7.8, Cliff delta = 0.27) and 6.5% (CI = 1.2–11.8, Cliff delta = 0.30), respectively, 32 hours after ML. No clear diurnal changes in CMJ or SJ occurred 8 hours following control; however, increases of 3.9% (CI = 2.9–9.2, Cliff delta = 0.26) and 4.5% (CI = 0.9–8.1, Cliff delta = 0.24), respectively, were observed after 32 hours. Although diurnal changes in saliva hormone concentration occurred (Cliff delta = 0.37–0.92), the influence of priming was unclear. Perceived "physical feeling" was greater 8 hours following HL (Cliff delta = 0.36) and 32 hours after ML and control (Cliff delta = 0.17–0.34). ***Conclusions:*** HL priming in the morning may result in small improvements in jump output and psychophysiological state in the afternoon. Similar improvements were observed in the afternoon the day after ML priming.

Keywords: resistance exercise, precompetition, athlete performance, athlete preparation, readiness to perform

Results of recent studies have shown strength, jumping, throwing, and sprinting performance to be improved within 48 hours of completing a bout of exercise.[1–4] This is referred to as priming exercise when prescribed with the aim of improving performance for a subsequent activity (eg, training or competition).[1,2,5,6] Among other methods,[1,7,8] various resistance exercise strategies have been shown to improve subsequent physical performance within this 48-hour period.[1,2,4] However, few studies have examined the effects of these methods. Improved understanding of the effects of resistance exercise priming may help practitioners appropriately prescribe these strategies in the applied environment.

Findings from previous studies have shown that high-load (HL) priming exercise can elicit small enhancements in physical capacities for up to 6 hours.[1,3,9] Cook et al[1] found performance improvements following priming, which involved 3 × 3 bench press and back squat at up to 3-repetition maximum (RM) load. Forty-meter sprint (1.3%, 95% confidence interval [CI] = 0.4%–1.5%), countermovement jump (CMJ) peak power (3.9%, CI = 2.5%–4.2%), and 3RM strength (≈ 4%, CI = 1.4%–4.7%) were all enhanced 6 hours after the priming activity. Comparable improvements in drop jump height[9] and ballistic throwing performance[3] have also been reported 4 to 6 hours following other HL priming activities. Collectively, available evidence suggests that HL priming can marginally improve aspects of physical performance within 6 hours. However, the performance response after 6 hours has not been examined and it is unclear if this strategy can also be effective after longer periods.

Recent findings have also shown that low- to moderate-load (30%–65% 1RM) priming may improve subsequent neuromuscular qualities for longer periods, up to 48 hours. Tsoukos et al[2] found CMJ height, reactive strength index, and isometric rate of force development were greater 24 and 48 hours following 5 × 4 squat jumps (SJ) at 40% 1RM. Performance improvements were greatest at 24 hours, when CMJ height was 5.1% (1.0%) greater than baseline, although remained 3% (0.7%) above baseline at 48 hours. This suggests that moderate-load (ML) priming strategies may be effective to improve performance on the day following its completion. However, research has yet to examine the effects of ML priming across the day of completion and the following day.

Considering the application of ML and HL priming methods in high-performance training programs,[6] it would be beneficial to examine the time course of responses following these strategies over the following days. Force–time analysis of CMJ has previously been used to describe neuromuscular changes following priming.[1,2,9] However, exploring SJ performance and additional variables (ie, CMJ phase analysis and strategy metrics)[10,11] will likely provide further insight into the effects on neuromuscular performance. Salivary testosterone (Sal-T) and cortisol (Sal-C) concentrations have also been used to indicate readiness to perform and have been shown to be influenced by priming exercise.[1,7,12] In addition, psychophysiological state has been found to be positively

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affected following sprinting priming[8] but has not been examined thoroughly following resistance-exercise priming methods. Accordingly, investigation of the neuromuscular, hormonal, and perceptual responses following priming may help improve understanding of the holistic effects of priming strategies on athlete performance.[13]

The primary aim of this study was to examine neuromuscular performance on the day of and the day following HL and ML priming exercise. A secondary aim was to additionally map the hormonal and perceptual changes over this period. It was hypothesized that HL priming would improve force–time characteristics in the afternoon of completion but not on the following day. In addition, it was anticipated that ML priming would be beneficial to performance on the day of and the day following completion.

Methods

Overview

A randomized crossover control trial with a repeated measures and counterbalanced design was used to examine the response following half squat priming interventions and control activities. Countermovement jump and SJ tests were conducted to assess neuromuscular performance changes prior to each activity and 5 minutes, 8 hours, 24 hours, and 32 hours after (Figure 1). Saliva was also collected at each time point for Sal-T and Sal-C analysis, in addition to a perceptual questionnaire. Following familiarization, anthropometric, and strength testing, participants were randomly assigned to complete control, HL priming, and ML priming in a counterbalanced order. Each activity (control, HL, and ML) was separated by a minimum of 3 and a maximum of 6 days (mean = 4.4 [1.5] d); participants attended the laboratory on a total of 14 occasions over a minimum of 15 days (Figure 1). Ethical approval was granted by The University of Queensland Institutional Research Ethics Committee.

Participants

In the study, 15 males (age = 23.8 [3.6] y, height = 181.6 [6.1] cm, body mass = 83.9 [10.2] kg, 1RM back squat strength = 139 [24.8] kg, 1.66 [0.23] kg BM^{-1}) provided informed written consent prior to participation. Inclusion criteria were males aged 18–35 years, minimum 12 months strength training experience, currently performing ≥1 lower-body strength session per week, and free from injury and illness.

Participants were instructed not to complete any lower-body resistance exercise in the 72 hours prior to 1RM or each testing session (Figure 1). They were also asked to avoid moderate- to high-intensity lower body exercise during this period and abstained from exercise the day before each trial until completing testing at the 32-hour time point. All permitted exercise completed was recorded in an activity diary and standardized across each trial. Participants also completed a 3-day nutritional intake diary during the first trial and were instructed to replicate their nutritional intake and timing for the remaining trials.

Strength Testing

At the start of the second familiarization session, 1RM back squat strength was assessed. The squat was completed to parallel depth, with the crease of the hips in line with the top of the knee musculature. A standardized warm-up, used in the experimental condition, was followed by a specific warm-up using previously described methods.[14] Squat depth was confirmed using 2-dimensional motion capture filmed 1 m to the right of participants at a height of 40 cm (Hudl Technique; Ubersense Inc, Chicago, IL). Internal knee and hip angles were recorded at the lowest point of the successful 1RM attempt (knee = 58.6° [4.7°], hip = 53.5° [3.7°]).

Experimental Activities

The standardized warm-up included 3-minute cycle ergometry at a rate of perceived exertion of 3 (RPE 0–10),[15] 5 × deep lunge rotations (each side), 10 m × high knee run, 10 m × heel flick run, 10 m × side skip (each direction), 10 × leg swings (each limb), 10 × supine hip bridge, 5 × deep squat, 3 × lateral squat (each side), 5 × pogo jumps, and 3 × CMJ at 80%, 90%, and 100% effort. During experimental trials, participants completed a specific warm-up of 1 × 5 half squats at 50% 1RM. Participants lowered to a depth of 90° internal knee flexion, with an elastic band attached to the

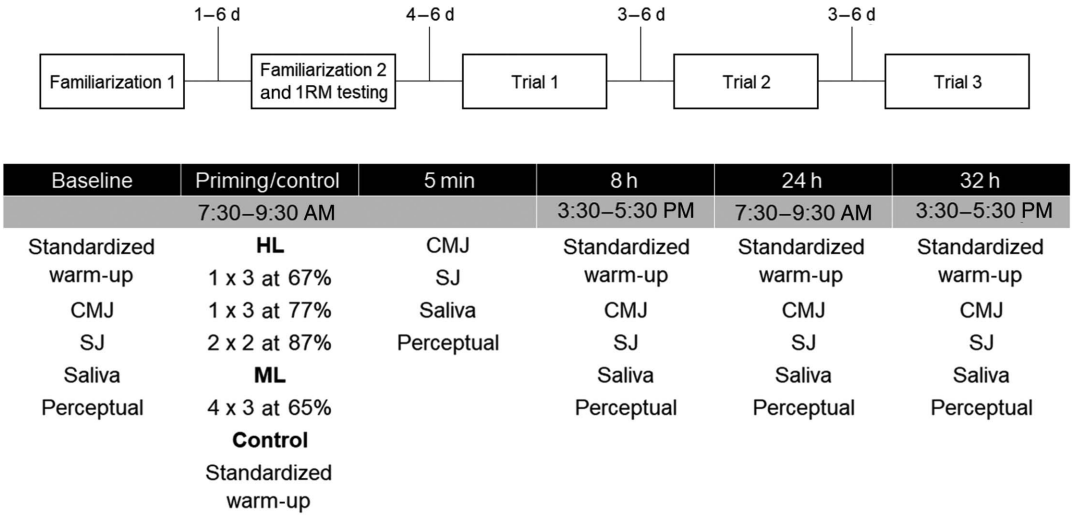

Figure 1 — Timeline of study procedures. 1RM testing indicates 1-repetition-maximum back-squat testing; CMJ, countermovement jump; HL, high-load; ML, moderate-load; Perceptual, perceptual questionnaire; Saliva, saliva collection for hormone analysis; SJ, squat jump.

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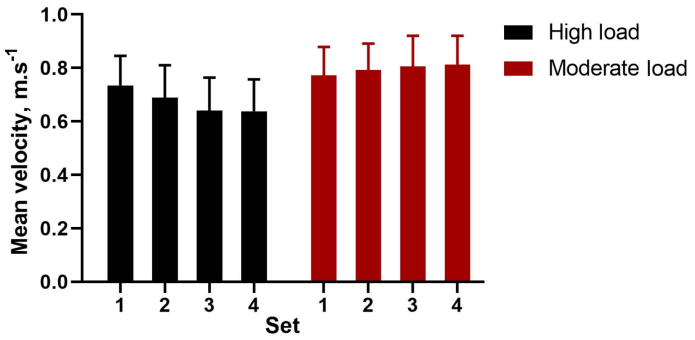

Figure 2 — Barbell mean (SD) velocity during HL and ML exercises. HL indicates high-load; ML, moderate-load.

squat rack for kinesthetic guidance. Joint angles for HL (knee = 87.7° [2.6°], hip = 84.4° [4.0°]) and ML (knee = 88.1° [2.7°], hip = 84.6° [3.3°]) half squats were recorded using aforementioned methods. The downward phase of the squat was completed in a controlled manner until contact with the elastic band before completing the upward phase at maximal intent of velocity. Toes remained in contact with the ground at all times and interset rest periods of 2 to 3 minutes were standardized across trials. Mean velocity of the half squat priming was recorded (Gymaware; Kinetic Performance Technology, Canberra, ACT, Australia) (Figure 2).

Neuromuscular Performance

Jump testing was completed on a Kistler force platform (Kistler, Ostfildern, Germany) that sampled at 1000 Hz from the vertical axis using native software (Kistler Bioware, version 5.1.3; Kistler). Testing involved 3 × CMJ followed by 3 × SJ separated by a 1-minute rest and 30 seconds between each jump. Participants completed jumps with hands akimbo and were verbally encouraged to aim for maximal height for each attempt. For SJ, participants lowered to a self-selected depth held for at least 3 seconds prior to jumping. If a countermovement presented on the force–time curve, the jump was repeated after an additional 30 seconds. 2 SJs presented a countermovement with unweighting of ≥10% from system mass and were removed from analysis as recommended by Sheppard and Doyle.[16] In these cases, the remaining 2 jumps collected were analyzed.

Phase analysis of CMJs involved dividing the jump into unweighting, braking, and propulsive phases using methods previously described.[17] Jump height, derived from take-off velocity (via the impulse–momentum relationship) was used as the criterion output variable in addition to causal (mean propulsive force, mean braking force, and countermovement displacement) and temporal (phase and total jump duration) variables.[10,11,18] Coefficient of variation (CV) of CMJ and SJ variables were between 2.6% and 9.2% and 2.3% and 5.8%, respectively. Intraclass correlation coefficients were between .85 and .93 for CMJ and .70 and .94 for SJ. Supplementary Material 1 (available online) contains detailed force–time analysis procedures, definitions, individual CV, and intraclass correlation coefficient for each variable.

Sal-T and Sal-C Measures

Saliva samples were collected using a synthetic swab (Salivette, Sarstedt, Germany) immediately after jump testing (Figure 1). Participants were instructed to avoid food and drink intake other than water for 30 minutes prior and mouth rinsed with water 10 minutes before unstimulated saliva collection for 60 seconds. Immediately following collection, samples were placed on ice and stored at −80°C within 30 minutes for later analysis.

On the day of analysis, sample tubes were thawed completely, centrifuged at 3300 × 10 minutes, and reweighed for saliva volume to calculate salivary rate. Sal-T and Sal-C were measured using immunoassay kits (Salimetrics, LLC, State College, PA). Testosterone was analyzed in duplicate (CV = 0.76%), and 1 in 5 cortisol samples was analyzed in duplicate (CV = 3.32%).

Perceptual Measures

After each jump test, participants responded to questions relating to their "physical feeling," "physical fatigue," "physical performance," and "effect of activity on performance" (available in Supplementary Material 2 [available online], which contains the full questionnaire). Responses were recorded on a 100-mm visual analog scale (VAS 0–100) and data were analyzed as raw values.

Statistical Analysis

For jump performance, the average of the 3 jumps completed during each test was analyzed. Data were processed using a customized MATLAB (MathWorks, Natick, MA) script. The distribution of each variable was examined with the Shapiro–Wilk normality test. Parametric data (SJ height and SJ duration) are presented as mean and 95% CI and compared with baseline using a 1-way repeated-measures analysis of variation with Bonferroni multiple comparison test. All remaining neuromuscular, hormone, and perceptual data were nonparametric. These are presented as median and 95% CIs with individual differences compared with baseline using the Friedman test and Dunn multiple comparison test. All statistical tests were analyzed in GraphPad Prism (GraphPad Software, San Diego, CA) and Microsoft Excel (Excel 2016; Microsoft, Redmond, WA). Current statistical approaches recommend that no threshold is set for significance.[19,20] Accordingly, *P* values are provided for all comparisons, although without associated terms or symbols describing statistically significant or nonsignificant change. For group effects, Cliff delta[21] was calculated, providing an effect size to identify thresholds for small (0.15–0.32), medium (0.33–0.46), and large (>0.46) changes. This calculates the proportion of overlap between samples in the groups, where a Cliff delta of 1 or −1 indicates no overlap and 0 represents complete overlap between groups. Change was considered to have occurred if 95% CI did not include the comparison value and Cliff delta met the threshold for a small group change. In addition, to identify individual changes in jump variables, the smallest worthwhile change was calculated using 0.2 × between-subject SD.[22] "Responding" and "non-responding" individuals were identified when individual change was smaller or greater than the smallest worthwhile change in each respective variable.

Results

Neuromuscular Performance

Baseline values are provided in Table S1 in Supplementary Material 1 (available online), while changes in CMJ and SJ variables are presented in Tables 1 and 2. Countermovement jump height was reduced by 3.7% and 5.1% 5 minutes following HL and ML, respectively (*P* = .007–.044, Cliff delta = 0.26–0.44).

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Table 1 Changes in Countermovement-Jump Metrics Following Control and Priming Activity as a Percentage Compared With Baseline

	Control (n = 15)				High-load priming (n = 15)				Moderate-load priming (n = 15)			
	%	95% CI	Cliff delta	*P*	%	95% CI	Cliff delta	*P*	%	95% CI	Cliff delta	*P*
Jump height												
Pre	100				100				100			
5 min post	98.7	96.5–100.5	–0.08	>.99	96.3	91.9–97.5	–0.26[S]	.007	94.9	93.1–97.4	–0.44[M]	.044
8 h post	103.7	100.5–105.3	0.12	.260	102.6	100.0–104.6	0.12	.663	103.1	100.1–104.3	0.16[S]	.816
24 h post	101.6	99.8–103.2	0.06	>.99	98.8	96.6–101.5	–0.02	.816	100.4	97.1–102.9	0.08	>.99
32 h post	103.9	102.9–109.2	0.26[S]	.001	104.8	100.0–106.6	0.28[S]	.259	106.1	102.1–107.8	0.27[S]	.022
Mean propulsive force												
Pre	100				100				100.0			
5 min post	98.5	98.9–100.9	0.00	>.99	99.8	97.9–101.1	–0.04	>.99	99.4	98.36–101.00	–0.02	>.99
8 h post	102.5	100.4–103.8	0.11	.032	102.3	100.9–104.4	0.16[S]	.016	101.2	99.86–103.10	0.07	.333
24 h post	99.2	98.9–101.3	–0.01	>.99	99.2	97.9–101.1	0.00	>.99	100.0	97.30–102.10	–0.06	>.99
32 h post	102.6	100.5–103.7	0.16[S]	.259	101.9	100.0–103.8	0.16[S]	.151	101.1	99.53–103.80	0.11	.816
Mean braking force												
Pre	100				100				100			
5 min post	99.5	98.3–102.0	0.00	>.99	98.2	97.2–104.4	–0.01	>.99	98.6	96.8–100.9	–0.06	>.99
8 h post	102.1	99.9–104.3	0.08	.333	102.7	101.8–109.9	0.16[S]	.113	100.9	99.4–105.5	0.01	>.99
24 h post	100.5	97.2–103.8	0.07	>.99	99.3	98.3–108.3	–0.04	>.99	104.1	100.0–107.6	0.08	.533
32 h post	101.6	97.9–105.4	0.08	>.99	103.4	102.1–111.9	0.19[S]	.032	105.6	101.1–109.4	0.14	.199
CM displacement												
Pre	100				100				100			
5 min post	99.3	95.5–100.9	–0.06	>.99	96.6	89.5–98.7	–0.23[S]	.061	95.9	93.9–97.9	–0.35[M]	.016
8 h post	101.7	99.1–104.6	0.08	.533	101.6	98.1–104.2	0.20[S]	.424	102.7	99.8–104.4	0.16[S]	.816
24 h post	100.9	99.5–104.0	0.06	>.99	98.3	94.8–101.1	–0.03	>.99	98.4	97.5–102.7	–0.03	>.99
32 h post	104.4	102.0–108.5	0.25[S]	.011	103.1	87.6–105.9	0.25[S]	.259	103.8	101.2–106.1	0.28[S]	.199
Brake phase time												
Pre	100				100				100			
5 min post	100.4	95.2–103.1	0.01	>.99	99.3	91.0–104.5	–0.03	>.99	98.0	93.7–101.2	–0.05	>.99
8 h post	95.7	93.5–101.9	–0.05	.533	96.9	85.7–99.9	–0.12	.424	95.6	91.0–101.8	–0.04	>.99
24 h post	101.3	94.2–104.9	–0.00	>.99	101.8	91.5–105.1	0.03	>.99	95.4	88.4–102.4	0.00	.533
32 h post	99.5	91.6–103.2	0.02	>.99	96.0	87.2–101.3	–0.13	.901	91.5	87.9–100.8	–0.05	.663
Propulsive phase time												
Pre	100				100				100			
5 min post	98.5	97.2–101.2	–0.05	>.99	100.4	96.1–101.2	0.00	>.99	98.4	95.7–100.4	–0.07	>.99
8 h post	96.3	94.8–100.6	–0.13	.151	98.0	94.1–100.2	–0.10	.227	98.0	95.4–101.4	–0.11	.663
24 h post	101.7	98.8–102.5	0.04	>.99	100.7	98.4–103.6	0.05	.816	99.0	95.7–105.1	–0.02	>.99
32 h post	98.6	96.4–101.5	–0.02	>.99	99.1	95.7–102.3	–0.03	>.99	99.9	95.3–102.9	–0.05	>.99

(continued)

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Table 1 *(continued)*

	Control (n = 15)				High-load priming (n = 15)				Moderate-load priming (n = 15)			
	%	95% CI	Cliff delta	*P*	%	95% CI	Cliff delta	*P*	%	95% CI	Cliff delta	*P*
Total jump time												
Pre	100				100				100			
5 min post	99.6	97.7–102.7	0.02	>.99	98.7	94.6–101.1	08	>.99	100.5	97.2–102.9	0.05	>.99
8 h post	99.4	96.9–102.1	–0.07	>.99	98.5	90.4–99.7	–0.16[S]	.663	98.3	94.2–101.6	–0.04	.816
24 h post	101.4	98.6–104.0	0.09	>.99	98.4	93.1–102.4	–0.13	>.99	98.1	93.3–103.2	–0.00	>.99
32 h post	99.0	96.1–105.0	0.05	>.99	97.0	89.9–99.6	–0.24[S]	.052	98.2	93.8–101.7	–0.05	.533

Abbreviation: CI, confidence interval; CM, center of mass.
[S] Small (0.15–0.32) change. [M] Medium (0.33–0.46) change.

Table 2 Changes in Squat-Jump Metrics Following Control and Priming Activity as a Percentage Compared With Baseline

	Control (n = 15)				High-load priming (n = 15)				Moderate-load priming (n = 15)			
	%	95% CI	Cliff delta	*P*	%	95% CI	Cliff delta	*P*	%	95% CI	Cliff delta	*P*
Jump height												
Pre	100				100				100			
5 min post	100.3	97.7–103.0	0.01	>.99	96.5	93.0–100.0	–0.19[S]	.170	95.8	93.2–98.4	–0.24[S]	.075
8 h post	102.9	99.5–106.3	0.09	.186	104.5	102.2–106.8	0.20[S]	.043	103.2	100.4–106.0	0.13	.252
24 h post	100.3	96.3–104.4	0.03	>.99	99.0	95.0–102.9	–0.06	>.99	101.4	97.1–105.6	0.06	>.99
32 h post	104.5	100.9–108.1	0.24[S]	.011	102.9	98.0–107.7	0.10	.395	106.5	101.2–111.8	0.30[S]	.002
Mean force												
Pre	100				100				100			
5 min post	99.5	99.0–100.4	–0.05	.663	97.7	96.9–100.1	–0.10	.816	98.7	97.1–101.5	–0.07	.816
8 h post	100.9	100.1–102.6	0.09	.993	101.2	100.3–102.7	0.12	.084	100.7	99.4–103.4	0.13	>.99
24 h post	98.4	96.7–99.0	–0.15[S]	.061	97.7	99.7–99.8	–0.11	.993	99.4	97.6–101.9	–0.03	>.99
32 h post	101.4	99.5–102.8	0.08	>.99	100.7	99.3–102.6	0.08	.259	103.1	100.3–105.0	0.16[S]	.084
Total jump time												
Pre	100				100				100			
5 min post	101.9	98.3–105.5	0.03	>.99	101.9	97.6–106.2	0.01	>.99	100.1	95.0–105.2	0.05	>.99
8 h post	98.8	95.8–101.8	–0.06	>.99	99.7	96.5–102.9	–0.04	>.99	98.7	94.5–102.9	–0.06	>.99
24 h post	106.3	101.8–110.9	0.24[S]	.005	104.5	100.5–109.6	0.17[S]	.085	101.1	96.1–106.1	0.03	>.99
32 h post	100.3	95.9–104.8	0.00	>.99	99.9	95.4–104.5	–0.07	>.99	96.8	92.1–101.6	–0.16[S]	.382

Abbreviation: CI, confidence interval.
[S] Small (0.15–0.32) change. [M] Medium (0.33–0.46) change. [L] Large (>0.46) changes.

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No change in jump height was observed between 8 and 24 hours, with the exception of a small increase 8 hours after ML (CI = 0.1–4.3, $P = .816$, Cliff delta = 0.16). However, at 32 hours, CMJ height was increased by 6.1% (CI = 2.1–7.8, $P = .022$, Cliff delta = 0.27) following ML and by 3.9% (CI = 2.9–9.2, $P = .001$, Cliff delta = 0.26) following control activities.

Squat jump height was also reduced 5 minutes after ML by 4.2% (CI = 1.6–6.8, $P = .075$, Cliff delta = 0.24). Following this, SJ height was improved by 4.5% (CI = 2.2–6.8, $P = .043$, Cliff delta = 0.20) 8 hours after HL, and by 6.5% (CI = 1.2–11.8, $P = .002$, Cliff delta = 0.30) and 4.5% (CI = 0.9–8.1, $P = .011$, Cliff delta = 0.24) 32 hours after ML and control, respectively. Figure 3 shows grouped and individual outputs in causal and temporal metrics for CMJ and SJ.

Salivary Hormone Measures

Salivary hormone data are presented in Table 3. Baseline Sal-T was greater in HL and ML compared with control ($P = .027$–$.080$, Cliff delta = 0.19–0.20). Moderate- to large-effect decreases in Sal-T occurred 8 and 32 hours following HL and ML ($P = .001$–$.008$, Cliff delta = 0.43–0.62). Only small effect decreases were observed after control activities, which occurred at 32 hours ($P > .99$, Cliff delta = 0.25). Sal-C was decreased in the afternoons, 8 and 32 hours after all activities ($P = .001$–$.037$, Cliff delta = 0.78–0.92) but not at 32 hours after control (CI = 80.0–108.6). Moderate effect increases in testosterone:control occurred 8 and 32 hours after HL and ML ($P = .005$–$.037$, Cliff delta = 0.37–0.41), with a large effect increase after control ($P < .001$, Cliff delta = 0.54–0.55).

Perceptual Measures

An increase in "physical feeling" occurred 5 minutes (58, CI = 51–71, $P = .333$) and 8 hours (59, CI = 53–73, $P = .052$) following HL compared with baseline (50, CI = 45–61, Cliff delta = 0.32–36). Small- and moderate-effect increases were reported 32 hours after ML (59, CI = 54–66, $P > .99$, Cliff delta = 0.17) and control (54, CI = 51–64, $P > .99$, Cliff delta = 0.34), respectively. No changes occurred at other times following any activity.

There was no change in perceived "physical fatigue" 8 hours following control or at 8 and 32 hours after ML. However, increases were observed following each activity at all other times (Cliff delta = 0.20–0.47), including moderate-effect increases 24 (37, CI = 21–43, $P = .019$, Cliff delta = 0.45), and 32 hours after HL (26, CI = 17–41, $P = .097$, Cliff delta = 0.40) compared with baseline (10, CI = 8–25).

There was a decrease in perceived "physical performance" 24 hours following HL (44, CI = 38–47, Cliff delta = 0.33) compared with baseline (50, CI = 44–53), although no other differences were found. Finally, there were no differences in perceived "effect of activity" between groups at any time. Individual perceptual data are presented in Figure 4.

Discussion

The primary aim of this study was to examine changes in jump performance over the 32 hours following HL and ML priming exercise. A secondary aim was to map changes in hormone and perceptual responses across this period. The main findings were that SJ height was improved 8 hours after HL, while SJ and CMJ height were improved 32 hours following ML. Aspects of psychophysiological state also appeared to be influenced at these times following priming activities; however, there was no clear effect of priming on hormone responses.

After HL priming, SJ height was improved by 4.5% at 8 hours compared with baseline. Although there was no increase in CMJ height, small changes to force–time characteristics occurred, including increases in mean propulsive and braking force. No clear changes were observed in duration of either phase; however, total jump time was reduced, which suggests that greater force was applied over less time in the CMJ. In addition, mean braking force of CMJ was increased in 7 positive responders (1RM = 129.6 [17.2] kg, 1.6 kg BM^{-1}) 8 hours following HL (+12.8% [3.8%]) compared with 8 nonresponders (–0.2% [2.1%]) (1RM = 146.9 [30.2] kg, 1.7 kg BM^{-1}) and 0 negative responders (Figure 3). Increases in braking force have been shown to occur after periods of strength–power training and are theorized to be central to adaptations in stretch-shortening-cycle (SSC) function.[18] Results of previous studies have shown positive effects of HL priming on CMJ peak power[1] and drop jump height[9] after 5 to 6 hours. However, no study has examined CMJ phase characteristics or SJ performance. Findings from the present study show that SJ output is improved and small changes to SSC function may also occur in the afternoon of completing HL priming exercise in the morning. Some improvements in CMJ force–time characteristics (increased braking force and decreased jump duration) were also present after 32 hours; however, SJ responses to HL priming appeared best after 8 hours. Despite this, it remains unclear if these changes influence other performance outcomes. Only one study has examined the effects of HL priming on the performance of specific sporting actions, such as sprinting.[1] As such, future research should now examine the effects of HL priming on broader tests of performance.

Following ML priming, CMJ and SJ height was increased by 6.1% and 6.5%, respectively, after 32 hours. Increases in jump output occurred in addition to small changes to force and strategy metrics. Squat jump height was improved with a small increase in mean force applied over the same duration. An increase in CMJ depth was also observed, while propulsive duration was unchanged. Accordingly, CMJ enhancements 32 hours following ML priming may be explained by improved utilization of the SSC, where force was applied over a greater distance in the same time. In addition, braking phase duration was decreased (–13.8% [5.7%]) in 9 positive responders (1RM = 1.7 [0.3] kg BM^{-1}) who concurrently displayed increased or unchanged jump height (+7.0% [4.3%]) and braking force (+10.1% [4.7%]). Two negative responders (1RM = 1.6 [0.4] kg BM^{-1}) displayed longer braking duration (+11.0% [2.0%]) and decreased braking force (–3.5% [1.8%]) but varied jump height response (+2.5% [6.4%]). Increases in braking force represent positive changes associated with improved SSC function[11,18] and the ability to apply force in limited time is essential to many sporting actions.[23,24] Consequently, jump output improvements and positive changes in force–time characteristics observed in this study suggest that ML priming may benefit physical performance capacities on the following day. Tsoukos et al[2] found similar qualities to be improved the day after squat jumps at 40% 1RM, reporting increased jump height, isometric rate of force development, and reactive strength index. Although most previous studies have examined the effects of HL priming,[1,3,9] these findings show that ML priming can also be an effective strategy to enhance neuromuscular function on the day following its completion.

Despite improvements at other times, CMJ and SJ height were not enhanced 24 hours after HL or ML. HL priming appeared to be

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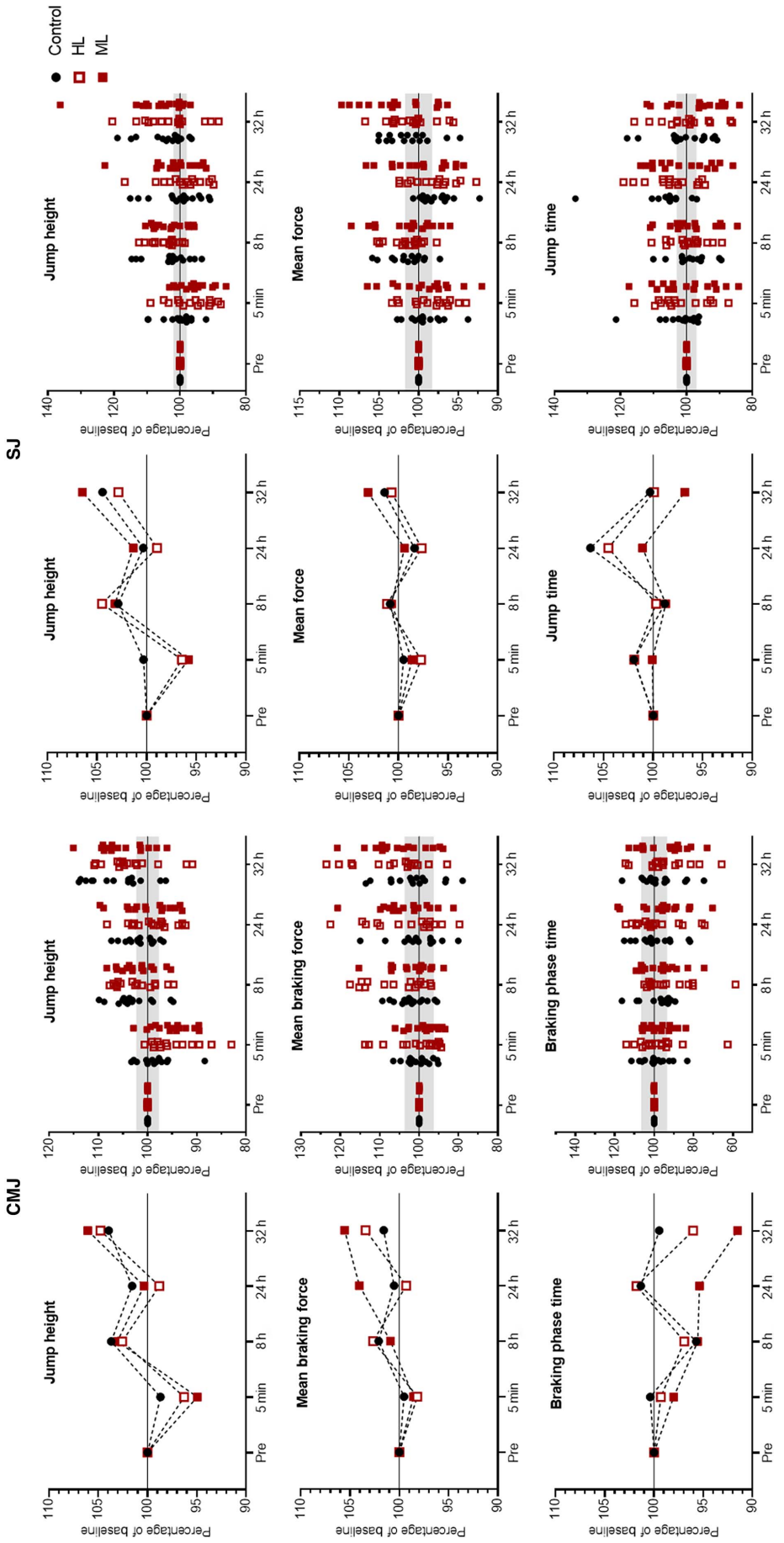

Figure 3 — Individual responses in CMJ and SJ output and associated changes in force and strategy metrics. Group changes (left) are presented as median/mean individual percentage of baseline. Individual data for each metric are presented to the right. The shaded area represents the smallest worthwhile change (0.2 × between-individuals SD). CMJ indicates countermovement jump; HL, high-load; ML, moderate-load; SJ, squat jump.

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Table 3 Salivary Hormone Data Presented as Percentage of Baseline Values

	Control				High-load priming				Moderate-load priming			
		95% CI	Cliff delta	*P*		95% CI	Cliff delta	*P*		95% CI	Cliff delta	*P*
Raw baseline												
Testosterone, pg·mL^{-1}	117.4	106.8–148.4			150.1	125.6–204.6	0.20^{S}	.080	137.5	126.5–168.3	0.19^{S}	.027
Cortisol, ng·mL^{-1}	8.9	7.3–13.1			8.2	7.1–12.0	0.18^{S}	.648	9.8	7.7–14.2	0.10	.893
Testosterone, %												
Baseline	100.0				100.0				100.0			
5 min post	103.1	97.9–111.2	0.05	>.99	95.5	86.1–100.8	–0.23^{S}	.547	93.7	88.2–105.8	–0.26^{S}	.690
8 h post	94.2	79.0–104.6	–0.32^{S}	>.99	84.6	66.3–88.8	–0.50^{L}	.001	71.2	68.5–80.4	–0.62^{L}	<.001
24 h post	108.3	100.3–125.0	0.14	.251	95.9	83.3–111.0	–0.23^{S}	>.99	94.2	86.8–105.2	–0.26^{S}	.859
32 h post	88.2	67.6–89.2	–0.25^{S}	>.99	78.4	67.6–89.2	–0.43^{M}	.001	75.7	68.8–88.8	–0.61^{L}	.008
Cortisol, %												
Baseline	100.0				100.0				100.0			
5 min post	78.4	70.0–99.7	–0.36^{M}	.329	72.4	66.4–83.8	–0.45^{M}	.189	76.3	64.0–94.1	–0.44^{M}	.189
8 h post	44.4	30.9–73.8	–0.82^{L}	.001	46.0	34.9–63.4	–0.88^{L}	<.001	45.2	29.6–72.1	–0.78^{L}	.002
24 h post	112.2	88.3–173.0	0.12	>.99	116.9	95.6–149.2	0.05	>.99	95.5	76.4–108.5	–0.25^{S}	>.99
32 h post	29.4	80.0–108.6	–0.85^{L}	<.001	41.3	32.1–63.5	–0.92^{L}	.001	39.6	32.1–63.5	–0.82^{L}	.001
Testosterone:control, %												
Baseline	100.0				100.0				100.0			
5 min post	126.6	114.5–142.5	0.14	.329	119.6	114.1–140.1	0.07	.189	131.2	103.1–165.5	0.12	.330
8 h post	187.2	157.1–142.5	0.55^{L}	.001	165.5	122.0–295.9	0.37^{M}	.012	174.8	121.0–353.0	0.38^{M}	.037
24 h post	96.8	75.4–136.8	–0.18^{S}	>.99	78.8	70.6–99.6	–0.38^{M}	.690	113.8	92.5–128.5	0.15^{S}	>.99
32 h post	269.2	205.1–430.9	0.54^{L}	<.001	157.8	139.5–269.0	0.39^{M}	.005	191.8	145.6–278.0	0.41^{M}	.005

Abbreviation: CI, confidence interval.

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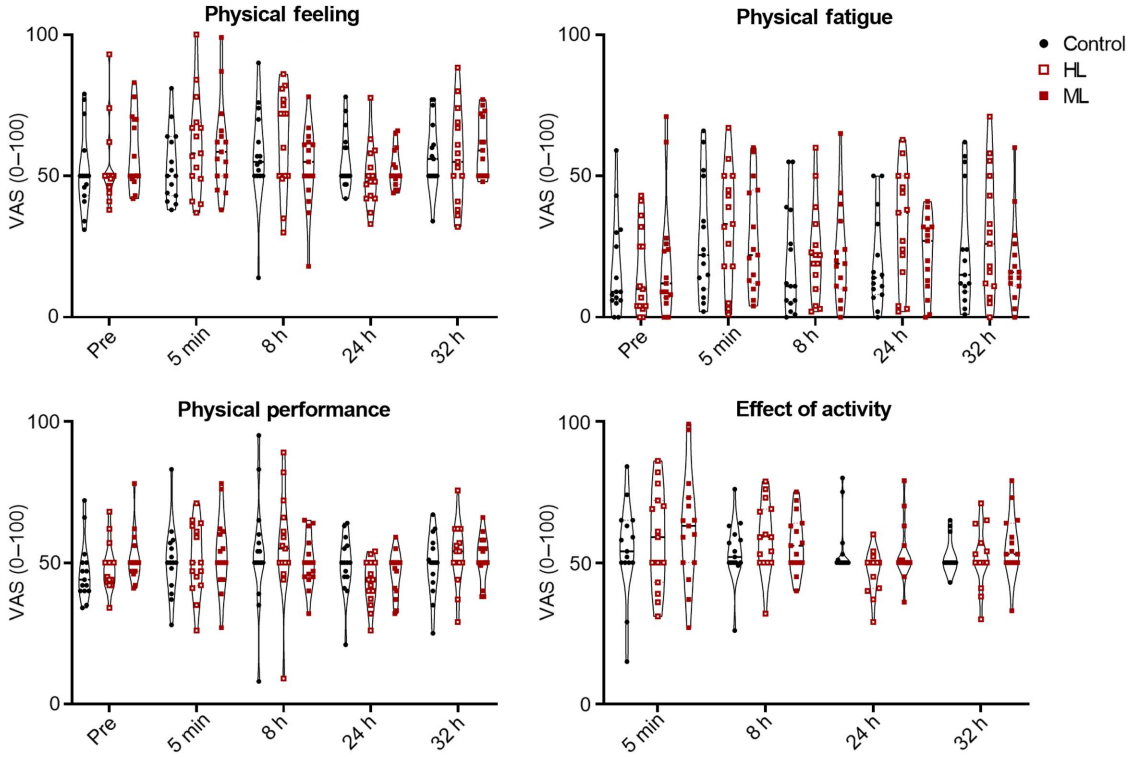

Figure 4 — Individual responses to perceptual questions following priming and control activities. Participants were asked, "physically, how do you feel right now? (0 = terrible, 100 = excellent)," "physically, how fatigued do you feel right now? (0 = not at all, 100 = maximally)," "compared to normal level of physical performance, how do you feel you could perform right now? (0 = well below normal, 100 = well above normal)," and "how do you feel the exercise session has directly prepared to perform physically right now? (0 = prepared me poorly, 100 = prepared me very well)." HL indicates high-load; ML, moderate-load; VAS, visual analog scale.

somewhat detrimental to performance at 24 hours, demonstrated by unchanged SJ height and force despite a 4.5% increase in jump time. Furthermore, this coincided with increased perceptions of physical fatigue and a decrease in perceived ability to perform. Previous research shows that fatigue responses occur the day following HL strength exercise[25]; however, this is a novel finding following HL priming. Although no improvements occurred 24 hours following ML, significant improvements have previously been shown 24 hours following SJ priming at 40% 1RM.[2] A possible explanation for this is that Tsoukos et al[2] assessed 24 hours performance in the afternoon (16:00–19:00), while the present study tested 24 hours performance in the morning (7:30–9:30). This suggests time of day is also an important consideration, where recovery processes may occur the morning following resistance priming exercise. As such, priming exercise the day before early morning activity does not appear optimal for performance outcomes and may be detrimental after HL priming.

No clear effect for diurnal changes were observed between morning (pre) and afternoon (8 h) jump performance following control. However, increases occurred after 32 hours when CMJ and SJ height improved by 3.9% and 4.5% above baseline, respectively. Increases in CMJ were accompanied by a 4.4% increase in countermovement depth, with minimal change to other metrics. This may explain the increased jump height, where the SSC is better utilized by allowing force to be applied over greater distance in the same time. In this study, jump testing procedures alone involved 32 minutes of dynamic warm-up and 28 maximal vertical jumps between baseline and 24-hour testing. Consequently, these activities may have resulted in the increased output at 32 hours, with no change observed at other times. Although it was not an aim of this study, the present data show that bouts of dynamic exercise incorporating maximal intensity may improve subsequent neuromuscular function. Considering these changes, and the application of unloaded jump priming in practice,[6] appropriately designed research examining the effects of these strategies may be warranted.

Analysis revealed no clear diurnal change in Sal-T following control, although large effect changes in Sal-C and testosterone: control occurred at 8 hours (Table 3). Baseline Sal-T was greater prior to HL and ML compared with control, suggesting that anticipatory increases in Sal-T may have occurred before the priming activities.[1] Consequently, there were generally greater decreases in Sal-T following priming and less substantial increases in testosterone:control. Regardless, similar responses would be observed between all groups if control baseline values were used in place of HL or ML baseline. This suggests that priming did not considerably influence hormone status across the period. These findings contrast studies by Cook et al[1] and Russell et al,[7] which both found priming exercise to attenuate diurnal decreases in Sal-T. This may be due to the increased volume of upper-body exercise prescribed by Russell et al[7] and inclusion of both upper- and lower-body exercise by Cook et al.[1] Nevertheless, findings from the present study indicate that Sal-T and Sal-C responses may not be central to improvements following HL or ML priming. Considering changes to jump height and other force–time metrics, central or peripheral neuromuscular mechanisms may explain these adaptations. Increased motor unit activation[9] and

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neural peripheral excitability occurs after resistance exercise[26,27] and may explain the neuromuscular responses following priming. Other changes including earlier activation patterns,[28] greater cross-bridge strength,[18] and changes in fascicle–tendon length relationships[29] have been suggested to drive adaptations in SSC function. It is possible that some of these changes also occur in the hours following priming exercise. However, further research in this area is warranted and may provide further direction regarding effective priming stimuli and the implications for athlete performance.

Some perceptual responses were similar to the observed changes in neuromuscular performance. Participants reported improvements in perceived "physical feeling" 8 hours following HL. In addition, a moderate-effect decrease in perceived "physical performance" and moderate increase in "physical fatigue" occurred at 24 hours after HL, resembling the SJ responses at these times. Small- and moderate-effect increases in "physical feeling" were also reported 32 hours following ML and control activities, respectively, which coincided with CMJ and SJ improvements. Marrier et al[8] also found that psychophysiological state was improved following sprint priming, but these measures have not been thoroughly explored in priming research. Due to the many factors that influence elite sporting performance,[13] consideration of both physical and psychophysiological responses may be insightful for coaches and athletes. Present findings suggest HL priming may benefit aspects of physical and psychophysiological state when prescribed in the morning before afternoon competition; however, they may have a negative influence the following morning. Small improvements to psychophysiological state may also occur in the afternoon, on the day after ML priming.

Previously, no study had examined the time course of responses over 2 days following 2 different priming exercise sessions. Findings support some previous research[1,3,9] that shows benefits of HL priming after 4 to 8 hours. A novel outcome from this study is that more substantial improvements occurred 32 hours following ML priming. Nonetheless, it should be considered that jump performance changes following priming are minor, particularly considering the measurement error of these variables. Although present findings identify changes during jumping, these may not necessarily translate into improved performance in other sporting actions, which warrants further investigation. In addition, due to the repeated bout effects of performance testing (pre, 5 min, 8 h, 24 h, and 32 h), the present experimental design did not allow for the comparison of responses between priming and control activities across these time points. The repeated physical testing appeared to influence performance 32 hours following control and may have also affected performance (positively or negatively) at other times or following HL and ML priming. Future research should examine specific measures of performance at one time point with highly trained male and female participants. In addition, similar research examining the time course of responses following other priming strategies (eg, sprinting, jumping, low load, and maximal-isometric resistance exercise) would be beneficial considering their application in practice.[6]

Practical Applications

Present findings suggest that HL and ML priming may both augment aspects of performance during subsequent activity. However, the time course of response appears to depend on the loading methods used. HL priming exercise appears to benefit neuromuscular and perceptual aspects of performance in the afternoon of completion. Conversely, ML priming was most effective to improve these qualities in the afternoon of the following day. These findings may assist coaches when planning resistance exercise during weekly preparation periods and may help to improve athlete readiness for training or competition. Coaches and researchers could also consider prescribing priming exercise using the corresponding barbell velocity of HL and ML priming (Figure 2) instead of relative 1RM loading.[30] Finally, improvements 32 hours after control suggest that repeated bouts of unloaded jumps at maximal intensity may also positively influence some neuromuscular qualities the day following their completion.

Conclusions

The HL and ML priming activities both appear to improve aspects of athlete performance in the subsequent 32 hours. Although there are many considerations for maximizing performance, practitioners should consider the short-term effects of HL and ML priming exercise when planning to augment athlete performance for training and competition.

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